1 March 1983

MILITARY SPECIFICATION

HEADS, MAGNETIC TAPE, INSTRUMENTATION, IRIG WIDE BAND GROUP II

(This specification is approved for use by all departments and agencies of the Department of Defense)

1.0. SCOPE

1.1. Scope. This specification describes the technical, performance and guarantee requirements for magnetic recording and reproducing heads (transducers) to be utilized for recording and reproducing information on magnetic tape in all instrumentation applications requiring wavelength response down to 1.524 micrometers (60 micro-inches), IRIG WB II. This specification shall be applicable to purchase specifications for replacement heads, as well as for heads supplied on new record/reproduce systems designed and manufactured after 1972.

1.2. Classification. Magnetic tape heads shall be of the following classes, which determine pole tip surface hardness and guaranteed life.

Type I, Sendust or equivalent. Hardness: Brinell, 495, or greater. Life: 1000 hours, operating.

Type II, Ferrite or equivalent. Hardness: Brinell, 750, Life: 2000 hours, operating.

1.2.1. Part Number. Specification part numbers for items described in this specification will be formulated as shown in paragraphs 6.5.

2.0. APPLICABLE DOCUMENTS

2.1. The current issue of the following documents form a part of this specification to the extent specified elsewhere in this specification.

2.2. Specifications.

Military

MIL-P-19834 Plate, Identification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: SM-ALC/MMEAM, McClellan AFB CA 95652, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.3. Standards.

Military

MIL-STD-105	Sampling Procedures and Tables Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standard General Requirements for Electronics Equipment
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests
MIL-STD-810	Environmental Test Methods

(Copies of military publications are available through the DOD single stock point. Commanding Officer, U.S. Naval Publications and Forms Center, Attn: NPFC 1032, 5801 Tabor Avenue, Philadelphia PA 19120)

2.4. Other Publications. The following documents from a part of this specification to the extent specified herein.

2.4.1. ANSI

ANSI B46.1-1978 Surface Texture.

(Copies of ANSI documents may be purchased from American National Standards Institute, Inc, 1430 Broadway, New York NY 10018)

2.4.2. ASTM

ASTM E104-76	Hardness Conversion Tables for Metals.
ASTM E10- 78	Brinell Hardness, Methods of Test for

(Copies may be obtained from American Society of Testing and Materials, 1916 Race St, Philadelphia PA 19103.)

2.4.3. IRIG

IRIG Standard 106-80	Telemetry Standards.
IRIG Standard 118-79	Test Methods for Telemetry Systems and Subsystems; Volume III, Test Methods for Recorder/Reproducer System and Magnetic Tape.

(Copies of IRIG documents are available for purchase from Secretariat, Telemetry Group, Inter-Range Instrumentation Group, Range Commander Council, White Sands Missile Range NM 88002.)

2.4.4. GSA-FSS

W-T-001553 Tape, Recording, Strumentation, Magnetic Oxide Coated

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(Copies of W-T-001553 may be obtained from General Services Administration, FSS (FMS) Washington DC 20406.)

2.4.5. Original Equipment Manufacturers. Individual model recorder/reproducer system performance specifications, alignment procedures and physical and electrical mounting interface requirements shall be used. (Individual system requirements are available from the various original equipment manufacturers or as specified in the contract.)

3.0. REQUIREMENTS

3.1. First Article. When specified (see 6.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.0 and 6.1).

3.1.1. Form, Fit and Function. All tape heads supplied under this specification shall, without exception, meet all mechanical and electrical form, fit and function parameters of the OEM equipment for which procured.

3.1.2. Product Qualification. All tape heads supplied under these specifications shall be products which have been tested and approved for listing on the applicable Qualified Products List (QPL) maintained by the United States Air Force at the Sacramento Air Logistics Center (SM-ALC/PMZ) McClellan AFB CA 95652.

3.1.2.1 Qualified Products Listing. The QPL shall consist of products which have been tested and have passed those qualification tests specified herein.

3.1.2.2. Qualification Product Listing. Products shall be submitted by prospective suppliers of magnetic tape heads to SM-ALC when so directed. Should the submitted products fail to meet the specification as defined herein, SM-ALC reserves the right to refuse to accept such products for additional qualification tests until satisfactory data and test results have been submitted indicating correction of product deficiencies. SM-ALC reserves the right to levy a charge to cover cost of testing for qualification of products.

3.1.2.3. Qualification Withdrawal. A supplier's qualified product listing may be withdrawn by SM-ALC in accordance with "Provisions Governing Application by Manufacturers for Inclusion on Federal QPLs (see 6.4).

3.1.2.4. Re-qualification. A supplier's product, once withdrawn from the QPL, shall not be accepted for re-qualification until satisfactory data and test results have been submitted to SM-ALC/PMZ by the supplier indicating correction of the product fault(s). SM-ALC reserves the right to levy a charge to cover cost of testing for re-qualification of products.

3.1.2.5. Process Change. Qualification and certification of a supplier's product under this specification, once established, applies only to those tape heads manufactured by the specific process in use at the time of qualification. All proposed process changes shall be reported to SM-ALC/PMZ with a statement by the supplier as to the general nature of the changes and the extent and effect of such changes on the delivered product. SM-ALC reserves the right to require six weeks notice from the supplier and sufficient samples of the new product for qualification tests prior to delivery of products manufactured under any such change in process. The foregoing requirement does not in any way relieve the manufacturer of delivery requirements, quality control and testing necessary to insure that all products delivered under this

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specification are equal or superior to those products originally submitted for qualification.

3.2. Design and Construction

3.2.1. General Design. Wideband Group II magnetic head assemblies which are provided in accordance with this specification shall operate at all speeds from 23.8 to 3048 Millimeters per second (mmps) (15/16 to 120 inches per second (ips) for at least 1000 hours (actual tape movement) or three years, whichever comes first. The heads shall be for use with magnetic tape configuration as specified by IRIG-106-80, Figure 6-1, Tape Geometry; track widths shall be in accordance with one of the configurations shown in Table I or as specified by the individual procurement.

3.2.2. Head Track Configuration. Reference IRIG-106-80. The tracks on a tape head shall be numbered as shown on Tables 6-1, 6-2, 6-3 and Appendix E. All test results from individual tracks shall be prepared in this format.

3.2.3. Record/Reproduce Bandwidths. Heads shall be designed for performance at the bandwidths described in Table II.

3.2.3.1. Wideband II. Direct record or reproduce of 1.524 micrometers (µm) (60 micro inch (µin)) wave length or 2 MHz at 3048 mmps (120 ips).

3.2.4. Heads. The heads provided shall meet the following requirements:

3.2.4.1. Head Placement. In conformance with IRIG-106-80, Section 6.3.2.1.

3.2.4.2. Head Stack Placement. In conformance with IRIG-106-80, Section 6.3.2.2.

3.2.4.3. Head Stack Numbering. Reference IRIG-106-80, Section 6.3.2.3.

3.2.4.4. Head and Stack Numbering. In conformance with IRIG-106-80, Section 6.3.2.4.

3.2.4.5. Head Stack Tilt. Reference IRIG-106-80, Figure 6.2. The plane tangent to the front surface of the head stack or any head stack mounted on an assemble at the centerline of the head gaps shall be perpendicular to the head-mounting place within ± 0.29 mrad (± 1 minute) of arc for heads provided in this specification.

3.2.4.6. Head Stack Gap Scatter. Reference IRIG-106-80, Figure 6-2. Gap scatter on specified heads shall be less than 0.00254 mm (0.0001 in) for all head configurations.

3.2.4.7. Mean Gap Azimuth Alignment. The mean gap azimuth alignment shall be perpendicular to the head-assembly mounting plate to within ± 0.29 mrad (± 1 minute) of arc.

3.2.4.8. Head Location in Assembly. Any head track Reference (Table I) in an assembly shall be located within ± 0.025 mm (± 0.001 in) of the nominal position required to match the track location as shown in IRIG-106-80, Figure 6-1 and Tables 6-1, 6-2, 6-3 and Tables E-1 and E-3.

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No.	Head Type/Assembly	No. Tracks	Tape Width	Track Width	Other Governing Specification
1	Record	7	12.7 ± 0.051mm (0.498 ± 0.002 in)	1.27 ± 0.051mm (0.05 ± 0.002 in)	IRIG-106-80, Table 6-1
2	Reproduce	7	"	"	"
3	R/R Assembly	7	"	"	"
4	Record	14	25.3 ± 0.051mm (0.998 ± 0.002 in)	"	IRIG-106-80, Table 6-2
5	Reproduce	14	"	"	"
6	R/R Assembly	14	"	"	"
7	Record	28	"	0.635 ± 0.025mm (0.025 ± 0.001 in)	IRIG-106-80, Table 6-3
8	Reproduce	28	"	"	"
9	R/R Assembly	28	"	"	"
10	Record	42	"	0.457 ± 0.025mm (0.018 ± 0.001 in)	IRIG-106-80, Table E-3
11	Reproduce	42	"	"	"
12	R/R Assembly	42	"	"	"
13	Record	14	12.7 ± 0.051mm (0.498 ± 0.002 in)	0.635 ± 0.025mm (0.025 ± 0.001 in)	IRIG-106-80, Table E-1
14	Reproduce	14	"	"	"
15	Record/Reproduce Assy	14	"	"	"

TABLE I

HEAD MECHANICAL REQUIREMENTS

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Tape Speed mmps	(ips)	± 3 db Pass Band (1) (Hz)	Record Bias (2) Set Frequency (kHz)	Record Level Set Frequency (kHz)
3048	120	400 - 2,000,000	2,000	200
1524	60	400 - 1,000,000	1,000	100
762	30	400 - 500,000	500	50
381	15	400 - 250,000	250	25
191	7 1/2	400 - 125,000	125	12.5
95	3 3/4	400 - 62,500	62.5	6.25
47.6	1 7/8	400 - 31,250	31.25	3.125
23.8	15/16	400 - 15,625	15.625	1.56

NOTE:

(1) Pass band response reference is the output at the record level set frequency.

(2) Using an input signal 5 to 6 db below normal record level, the record bias current is adjusted for maximum reproduce output and then increased until an output of 2 db below the maximum value is obtained.

TABLE II
TEST FREQUENCY RECORD/REPRODUCE
PERFORMANCE PARAMETERS

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3.2.4.9. Angular Gap Alignment. The head stacks shall be assembled on the mounting assembly such that the center line of the gaps of each head stack is the perpendicular bisector of the tape wrap angle (given by the OEM) within ± 36 minutes of arc.

3.2.5. Electrical Requirements.

3.2.5.1. Head Assemblies. The record and reproduce head assemblies supplied under this specification shall be equivalent to or exceed the electrical performance requirements originally placed on the recording system by the original equipment manufacturer (OEM).

3.2.5.1.1. Grounding. All housings, (cores), and shields shall have less than 1 ohm resistance to the mounting plate and shall meet OEM requirements.

3.2.5.1.2. Connectors. All connectors shall meet the OEM requirements both physically and mechanically and shall meet the requirements of MIL-STD-454, Requirement 10, if applicable.

3.2.5.1.3. Cabling and Wires. The wiring used shall meet or exceed the OEM requirements and Requirements 17, 65, 66, 69 and 71, MIL-STD-454, both physically and electrically.

3.2.5.2. Record Head. Manufacturer performance data shall be supplied with each head.

3.2.5.2.1. Bias Current. The bias current requirements shall not exceed OEM requirements when set up at all points given in Table II.

3.2.5.2.2. Bias Voltage. The peak-to-peak bias voltage necessary to achieve the required bias current shall not exceed OEM requirements when set up at all points given in Table II.

3.2.5.2.3. Record (data) Current. The record current shall not exceed OEM requirements when set up at all points given in Table II.

3.2.5.2.4. Record Head Polarity. In conformance with IRIG-106-80, Page 6-9, Section 6.3.3.1.

3.2.5.2.5. Record Head Track-to-Output Flux Variation. The level of flux at the gap when recording a constant analog signal at record level set to the frequencies described in Table II shall not cause a variation in the reproduce signal between individual tracks of more than a 6 db spread at 3048 mmps (120 ips). Record head pre-emphasis and impedance must meet the OEM requirements. For this test the vendor is to utilize a "reference" reproduce head of known quality, where variations within this "reference" head are known and documented.

3.2.5.2.6. Head Resonance. The head assembly shall resonate at a frequency equal to or greater than the requirements of the OEM.

3.2.5.2.7. Core and Tip Saturation. Head core or core tip saturation shall not occur prior to tape saturation when using a tape purchased under WT-0001553 type E2 oxide.

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3.2.5.3. Reproduce Head. The following parameters shall be certified by the manufacturer. In addition, manufacturer performance data of paragraphs 3.2.5.3.1, 3.2.5.3.2, and 3.2.5.3.3 shall be supplied with each head when specified in 6.1.e.

3.2.5.3.1. Reproduce Voltage Output. The reproduce output from each track at peak and band edge when terminated according to 3.2.5.3.4 shall be within the OEM specified value of voltage subject to the variation defined in 3.2.5.3.2.

3.2.5.3.2. Reproduce Voltage Output Variation. The maximum variation in output voltage from one track to any other when terminated according to 3.2.5.3.4 may not be more than 6 db at peak frequency at the maximum and minimum speed for which the head is being procured.

3.2.5.3.3. Slot Signal-to-Noise Ratio. The ratio of RMS signal to RMS noise on any track shall not be less than that specified by the OER, when terminated according to 3.2.5.3.4.

3.2.5.3.4. Termination Impedance. The head shall be designed to operate with each track terminated by the OEM required impedance.

3.2.5.3.5. Crosstalk. Crosstalk herein is defined as the RMS reproduce voltage of the track under test at band edge as a result of all other tracks having been recorded at a normal record level. The RMS voltage difference between tracks shall be equal to or greater than that specified by the OEM for all tracks of the head assembly.

3.2.5.3.6. Crosscoupling. Crosscoupling herein is defined as the reproduce voltage of the reproduce tracks with normal record current across the frequency spectrum of usage applied to all record tracks on all record heads, with no tape on the heads. This voltage shall be no greater than that specified by the OEM on all tracks of the head assembly.

3.2.5.3.7. Reproduce Head Polarity. In conformance with IRIG-106-80, Page 6-9, Section 6.3.3.2.

3.2.6. Record/Reproduce Electronic Parameters. The bias frequency for record should be equal to or greater than 3.5 times upper band edge frequency and shall result in response of recorded data as a function of tape speed as shown in Table II. In measuring this response, signals throughout the specified pass band are recorded at normal record level and the reproduce output signal levels are referenced to the reproduce output at the record level test frequency.

3.2.7. Mechanical Requirements. The following parameters shall be tested or certified as required by the specification, by the manufacturer. In addition, performance data shall be supplied with the first article (see Section 6.1).

3.2.7.1. Head Assembly. The outline profile, detail dimensions, mounting configuration including all angles shall be according to OEM specifications. Measurements shall be per 4.7.2.1 and 4.7.2.2.

3.2.7.2. Head Surface. The surface of all magnetic materials, and their surrounding metallic structure contacting the tape shall have a surface finish of no more than .20 m (3 in) (peak-to-valley as measured by ANSI B-46.1. Such measurement shall be made prior to passing tape over the head. Any epoxy bond lines in the contact area of the head face are to be no more than .051 mm (.002 in) wide. Further, the surface of these epoxy bond lines shall always be

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below the lowest metallic face surface of the head, but their depth shall not exceed 1.27 m (50 in). The passage of 2804 m (9,200 ft) of tape W-T-001553) over the head in normal environment (see 3.5.2 and 3.5.4) shall cause no physical damage to the surface of the tape.

3.2.7.3. Mechanical Interface. All mechanical interfaces shall be fabricated to insure that the head assembly is interchangeable with and meets the form, fit and function of the OEM head as per the GFE sample of the OEM head specification.

3.2.7.4. Record Head.

3.2.7.4.1. Gap Length. The gap length shall be 2.29 ± 0.25 m (90 ± 10 in) except where this requirement will make impossible to meet OEM specifications. Measurements shall be per 4.7.2.1.

3.2.7.4.2. Gap Depth. The gap depth shall be measureable externally to the head and shall not be less than 0.051 mm (0.002) unless OEM specifications dictate lesser depth. Measurements shall be per 4.7.2.1.

3.2.7.4.3. Azimuth Adjustment. An azimuth adjustment shall be provided if required by OEM specifications.

3.2.7.5. Reproduce Head.

3.2.7.5.1. Gap Length. The reproducer gap length shall meet OEM specifications. Measurement shall be per 4.7.2.1.

3.2.7.5.2. Gap Depth. The reproduce head gap depth shall be measureable externally to the head and shall not be less than 0.051 mm (0.002 in) for Type I heads and 0.038 mm (0.0015 in) for Type II heads unless OEM specification dictate a lesser depth. Measurement shall be per 4.7.2.1.

3.2.7.5.3. Azimuth Adjustment. An azimuth adjustment shall be provided if required by OEM specifications.

3.2.7.5.4. Azimuth Reproduce Output. The reproduce output of each track at 2.0 MHz at the optimum azimuth adjustment position for each head assembly shall be within 2 db of the output at the optimum position for each individual track.

3.2.7.5.5. Center Tap. The windings shall be tapped at the center of the reproduce pole winding within 1/2 turn of the center of the winding if a center tap is required by OEM requirements.

3.2.8. Material Requirements.

3.2.8.1. Head Assemblies. Brass shall not be used in the area of the surface contacting the tape. Non-metallic materials (plastic or epoxy used in constructing the head) shall be kept to a minimum and shall not contribute to the buildup of deposits on the head.

3.2.8.2. Pole Tips. The cores (poles) of each track of both record and reproduce heads shall either be made, or tipped with hard materials as defined herein. Mu-metal tips are not permitted. Reference is made to ASTM Publications E10-73 and E104-76 which provide numerous methods for measuring material hardness.

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3.2.8.2.1. Type I Tips. Type I material for the purposes of this specification is defined as a generic metal alloy typically using various combinations of aluminum, iron and silicon as well as other materials. Proprietary varieties of materials, such as Sendust, under manufacturer unique trademarks or names, are acceptable providing they meet this specification. The tips shall be fabricated from material meeting one of the requirements of Table III.

3.2.8.2.2. Type II Tips. Type II material for the purposes of this specification is defined as generic metal typically various combinations of ferrite. Proprietary varieties of Type II material, headcores, and tips, under unique trademarks or names are acceptable providing they meet this specification. The tips shall be fabricated from material meeting one of the requirements of Table III.

Table III
MATERIAL HARDNESS TEST VALUES

Hardness Units/Methods	HEAD TYPE	
Reference ASTM E10-73	Type I Test Value	Type II Test Value
Brinell	495	750
Rockwell C	48	62
Knoops	510	776
Vickers	505	800

3.2.8.3. Intertrack Materials. The intertrack materials (shields, epoxy, bracket) shall be selected such that over the life of the head, oxide, binder, or other contaminant build up will be minimized. These materials shall have a wear rate equal to or greater than the pole tips to ensure positive tape to pole tip contact.

3.2.8.4. Gap Material. The gap material shall be selected to insure "definition" over the life of the head. Definition includes lack of smearing in tipped heads and erosion in ferrite heads under normal environmental conditions (ref 3.5.2 and 3.5.4). The head contour shall be selected to insure positive tape to head contact and minimize flying effects.

3.2.8.5. Fasteners. All fasteners, screws, threads, receptacles and interfaces shall meet OEM requirements, both physically and mechanically, and shall meet the requirements of MIL-STD-454, Requirement 12, if applicable.

3.2.9. Environmental Requirements. The heads specified within this specification shall operate within the OEM recorder system environmental specifications.

3.3. Active Electronics in Head Assembly. Any active electronics which are required as a part of the head assembly shall be provided and tested to meet OEM specifications.

3.3.1. Electrical Interfaces. Head wiring, cables (wires), connectors, printed wiring and external electronic boards and cards, where required, shall be designed and constructed in accordance with OEM requirements.

3.4. Marking and Identification. The head stack or head stack assembly shall be identified and marked with an environmentally protected label. When

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practicable, the label shall be visible to the tape transport operator when the heads are mounted on the transport. The following information shall be contained on the label(s):

1. Manufacturer Name or Logo

2. Part Number - Abbreviation: PN

3. Serial Number - Abbreviation: SN

4. Warranty_____Hours/_____Years

5. Date Installed_____

6. Head Meter Reading_____

The spaces following item numbers 5 and 6 above shall be of label material which allows for field identification using a ballpoint pen and shall be of sufficient size to record the required information.

3.5. Environmental Operating Range.

3.5.1. Severe Operating Temperature. + 1°C to 50°C ± 5°C.

3.5.2. Normal Operating Temperature. + 25°C ± 5°C.

3.5.3. Severe Operating Relative Humidity. 0% to 99% Non-condensing.

3.5.4. Normal Operating Relative Humidity. 40% to 60%.

3.5.5. Thermal Rate of Change. The head stack assembly shall operate within all requirements of this specification and shall not be damaged when subjected to a thermal rate of change of 1°C per minute throughout the entire operating range unless the recorder or head require a greater rate of change as specified by the OEM.

3.6. Workmanship. Workmanship shall be of a quality consistent with industry standards and practices. All finished surfaces shall be protected against corrosion or damage during manufacturing and prior to delivery. All materials, which may adversely affect the endurance or wear resistance of the tape head assembly, shall be of at least equal quality and condition and be free from defects when referenced to OEM specification. MIL-STD-454, Requirement 9, shall apply.

3.7. Reliability. Tape heads shall meet the following MTBF requirements:

Type of Tape Head	Lower Test MTBF	Upper Test MTBF
Type I	1000 hrs.	3000 hrs.
Type II	2000 hrs.	6000 hrs.

3.7.1. Definition of Failure. Tape heads will be considered to have failed any time they do not operate in accordance with Section 3.0, Requirements, or the following wear limits have been exceeded:

		Wear Limit
(a)	Type I	100 micro inches
(b)	Type II	35 micro inches

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3.7.2. Definition. Note that Lower and Upper Test MTBF are as defined in MIL-STD-781.

3.8. Preventive Maintenance. The only periodic maintenance permitted shall be cleaning of tape heads once every 400 hours of operation.

4.0. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for Inspection. Unless otherwise specified in the contract, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Quality Assurance Terms and Definitions. Quality assurance terms and definitions shall be as specified by MIL-STD-109.

4.2. Certificate of Compliance. When a certificate of compliance is furnished for any requirement specified herein, the Government will have the right to check test items to determine the validity of the certification. Where defects or inferior quality is evident, and the Government deems a material analysis necessary, the contractor shall be requested to submit samples to the contracting officer for analysis.

4.3. Classification of Inspections. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.5).

b. Quality conformance inspection (see 4.6).

4.4. Test Preparation.

4.4.1. Atmospheric. Unless otherwise specified, the tests shall be performed at 23°C ± 1.0°C and a maximum relative humidity of 50 percent.

4.4.2. Test Tapes. Prepare a quantity of ten 2804m (9200') or twenty 1463m (4800') sample tapes procured and prepared in compliance with W-T-001553, paragraph 4.4.1.

4.4.3. Head Profile. Prepare a profile of the recording surface at the gap of the new heads to be subjected to test using a profilometer with a resolution of at least ± 0.051 um (± 2 uin).

4.4.4. Test Set Up. Mount the heads on a tape transport of the same model as that for which the heads are intended in the end user application. Prepare the tape transport for the test by complete mechanical and electronic alignment per original equipment manufacturer procedures. Concern shall be taken to insure correct tape tension per OEM requirements.

4.5. First Article Inspection. Unless otherwise specified in the contract or purchase order, the first article inspection shall consist of all the tests as specified by Table IV.

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4.5.1. Samples. A minimum of two head assemblies shall be subjected to first article inspection. The assemblies shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production.

4.6. Quality Conformance Inspection. Quality conformance inspection shall consist of sampling inspections.

4.6.1. Sampling Inspection. The tape heads shall be subjected to the sampling inspection specified in Table V.

4.6.1.1. Inspection Lot. Unless otherwise specified by the contracting officer, inspecting lot size, formation and presentation of lots shall be in accordance with Submission of Product and Drawing of Samples specified in MIL-STD-105.

4.7. Methods of Examination and Test.

4.7.1. Reliability. Requirements of paragraph 3.7 shall be demonstrated in accordance with MIL-STD-781, Test Method XVII C. Tape speed during the test shall be 3048 mmps (120 ips), the test environment shall be in accordance with paragraphs 3.5.2 and 3.5.4 of this specification, and accept-reject criteria shall be as specified in the Test Plan. During the test, the tape head may be cleaned as specified in paragraph 3.8. No other preventive maintenance tasks shall be allowed. All failures occurring during this demonstration shall be recorded, appropriate repairs performed, and the repaired tape heads returned to service. If the tape head is not reparable, a new tape head shall be substituted for the failed unit. Note that the acquisition activity shall be the final authority in determining the chargeability of all failures.

4.7.2. Examination.

4.7.2.1. Optical Measurements. Using a microscope of at least 100 power and a micrometer adjustable stand with resolution to 0.0254 mm (.001 inch) measure and record the width, height or length of the required dimensions.

4.7.2.2. Optical Angular Measurements. Optical angular measurements using a microscope with at least 100 power and an angular measuring base with a resolution of at least one second, measure and record the angle of the required measurements.

4.7.2.3. Head Polarity. Test procedures in accordance with IRIG Document 118-79, Volume III, Chapter I, Section 1.1, shall be used.

4.7.2.4. Record/Reproduce System Test. Test procedure in accordance with IRIG Document 118-79, Volume III, Chapter IV, Sections 4.1 and 4.2 shall be used.

4.7.2.5. Grounding. Using low impedance ohmeter, measure and record the resistances of all housing and shields per paragraph 3.2.5.1.1.

4.7.2.6. Bias Current. Using the procedures described by 4.7.2.4, measure the bias current on all tracks with an oscilloscope clamp-on ammeter probe with conditions described by Section 4.1.3.3.1, Volume III, IRIG Document 118-79.

4.7.2.7. Bias Voltage. Using an oscilloscope, measure the voltage on all

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tracks measured in test procedure 4.7.2.6.

TABLE IV First Article Inspection

Requirement	Requirement Paragraph	Test Paragraph
Article	3.1	4.5
General Design	3.2.1	4.7.2.1
Head Track Numbering	3.2.2	3.2.4
Record/Reproduce Bandwidths	3.2.3	3.2.4
Head Placement	3.2.4.1	4.6
Head Stack Assembly Placement	3.2.4.2	4.7.2.1
Head Stack Assembly Numbering	3.2.4.4	N/A
Head and Stack Numbering	3.2.4.4	N/A
Head Stack and Assembly Tilt	3.2.4.5	4.7.2.2
Head Stack Gap Scatter	3.2.4.6	4.7.2.1
Mean Gap Azimuth Alignment	3.2.4.7	4.7.2.2
Head Location in Assembly	3.2.4.8	4.7.2.1
Angular Gap Alignment	3.2.4.9	4.7.2.2
Head Assemblies	3.2.5.1	4.7.2.4 OEM TESTS
Grounding	3.2.5.1.1	4.7.2.5
Connectors	3.2.5.1.2	4.7.2.19
Cabling Wires	3.2.5.1.3	4.7.2.19
Record Head	3.2.5.2	OEM Tests
Bias Current	3.2.5.2.1	4.7.2.6
Bias Voltage	3.2.5.2.2	4.7.2.7
Record Current	3.2.5.2.3	4.7.2.8
Record Head Polarity	3.2.5.2.4	4.7.2.3
Record Head Output Variation	3.2.5.2.5	4.7.2.9
Head Resonance	3.2.5.2.6	4.7.2.19
Core & Tip	3.2.5.2.7	4.7.2.19
Reproduce Head	3.2.5.3	OEM Test
Reproduce Voltage Output	3.2.5.3.1	4.7.2.10
Reproduce Voltage Output Variation	3.2.5.3.2	4.7.2.10
Slot Signal-to-Noise Ratio	3.2.5.3.3	4.7.2.11
Termination Impedance	3.2.5.3.4	4.7.2.19
Crosstalk	3.2.5.3.5	4.7.2.12
Crosscoupling	3.2.5.3.6	4.7.2.19
Reproduce Head Polarity	3.2.5.3.7	4.7.2.3
Record/Reproduce Electronic	3.2.6	4.4.4 and 4.7.2.4
Mechanical Requirements	3.2.7	OEM Tests
Head Assembly	3.2.7.1	4.7.2.1; 4.7.2.2
Head Surface	3.2.7.2	4.7.2.13
Head Assembly	3.2.7.3	4.7.2.14
Gap Length	3.2.7.4.1	4.7.2.1
Gap Depth	3.2.7.4.2	4.7.2.1
Azimuth Adjustment	3.2.7.4.3	4.7.2.14
Gap Length	3.2.7.5.1	4.7.2.1
Gap Depth	3.2.7.5.3	4.7.2.14
Azimuth Reproduce Output Rqmt	3.2.7.5.4	4.7.2.15
Head Assemblies	3.2.8.1	4.7.2.19
Pole Tips	3.2.8.2	4.7.2.19

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Table IV First Article Inspection (Contd)

Type I Tips	3.2.8.2.1	4.7.2.16
Type II Tips	3.2.8.2.2	4.7.2.16
Intertrack Materials	3.2.8.3	4.7.2.19
Gap Materials	3.2.8.4	4.7.2.19
Fasteners	3.2.8.5	4.7.2.19
Environmental Requirements	3.2.9	OEM Tests
Active Electronics	3.3	4.7.2.17
Electrical Interface	3.3.1	4.7.2.19
Marking Identification	3.4	4.7.2.14
Severe Operating Temperature	3.5.1	4.7.2.19
Severe Operating Relative Humidity	3.5.3	4.7.2.19
Thermal Rate of Change	3.5.5	4.7.2.18
Workmanship	3.6	4.7.2.19
Head Life	3.7	4.7.1
Preventive Maintenance	3.8	4.7.1

Table V Sampling Inspection

Req. Par.	Test Par.	Test or Certify	Percent of Test Articles to be Tested
3.2.1	4.7.2.1	T	20
3.2.4.2	4.7.2.1	T	20
3.2.4.5	4.7.2.2	C	20
3.2.4.6	4.7.2.1	T	20
3.2.4.7	4.7.2.2	T	20
3.2.4.8	4.7.2.1	T	20
3.2.4.9	4.7.2.2	C	
3.2.5.1	4.7.2.4	T	100
3.2.5.1.1	4.7.2.5	C	
3.2.5.1.2		C	
3.2.5.1.3		C	
3.2.5.2.1	4.7.2.6	T	100
3.2.5.2.2	4.7.2.7	C	
3.2.5.2.3	4.7.2.8	T	100
3.2.5.2.4	4.7.2.3	C	
3.2.5.2.5	4.7.2.9	T	100
3.2.5.2.6		C	
3.2.5.2.7		C	
3.2.5.3.1	4.7.2.10	T	100
3.2.5.3.2	4.7.2.10	T	10
3.2.5.3.3	4.7.2.11	T	20
3.2.5.3.4		C	
3.2.5.3.5	4.7.2.12	T	20
3.2.5.3.6		C	
3.2.5.3.7	4.7.2.3	C	
3.2.6	4.7.2.4	T	100
3.2.7.1	4.7.2.1 & 4.7.2.2	C	
3.2.7.2	4.7.2.13	C	
3.2.7.3	4.7.2.14	T	20
3.2.7.4.1	4.7.2.1	C	
3.2.7.4.2	4.7.2.1	T	20
3.2.7.4.3	4.7.2.14	T	100
3.2.7.5.1	4.7.2.1	C	

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Table V Sampling Inspection (Contd)

3.2.7.5.2	4.7.2.1	T	20
3.2.7.5.3	4.7.2.14	T	100
3.2.7.5.4	4.7.2.15	C	
3.2.7.5.5		C	
3.2.8.1		C	
3.2.8.2		C	
3.2.8.2.1	4.7.2.16	C	
3.2.8.2.2	4.7.2.16	C	
3.2.8.3		C	
3.2.8.4		C	
3.2.8.5		C	
3.3	4.7.2.17	C	
3.3.1		C	
3.4	4.7.2.14	C	100
3.5.5	4.7.2.18	C	
3.6		C	
3.7	4.7.1	C	

4.7.2.8. Record Current. Using the procedures described by 4.7.2.4, measure the record current on all tracks with an oscilloscope clamp-on ammeter probe, with conditions described by Section 4.1.3.3.2, Volume III, IRIG Document 118-79.

4.7.2.9. Record Head Output Variation. Record a tape using procedures described in Section 4.1, Volume III, IRIG Document 118-79. Reproduce the tape using procedures described in Section 4.2, Volume III, IRIG Document 118-79. Log the reproduce voltage per 4.2.3.4, Volume III, IRIG 118-79.

4.7.2.10. Reproduce Voltage Output. Record and reproduce a tape using procedures described in Section 4.2., Volume III, IRIG Document 118-79. Reproduce the tape and log the reproduce head voltage output at peak and bandedge from each track.

4.7.2.11. Slot Signal-to-Noise Ratio. Using the procedures described in Section 4.3, Volume III, IRIG Document 118-79, and substituting a wave analyzer having a three kHz slot bandwidth at the frequencies specified by the OEM for the bandpass filter specified in Section 4.3.3.1, IRIG 118-79. Measure and log the slot signal to noise ratio for each track.

4.7.2.12. Crosstalk. Record a 2.0 MHz data signal on the track under test at 3048 mmps (120 ips) per procedures described in paragraph 4.7.2.4. Rewind tape and log reproduce data signal level of the track under test per paragraph 4.7.2.4. Degauss the tape and record a 2.0 MHz data signal on all tracks except the track under test per 4.7.2.4. Rewind the tape and measure the reproduce level of the track under test using a wave analyzer having a 3kHz slot bandwidth at 2.0 MHz. Repeat procedure for all tracks in head assembly.

4.7.2.13. Head Surface.

4.7.2.13.1. Using the procedures described in ANSI-B-46.1-1978, prepare a surface profile of the active head area. Log the differences between peak and valley excursion.

4.7.2.13.2. After passage of 2804m (9200 ft) of tape 120 ips (3048 mmps), examine the tape at 500 feet in from both ends and at 1463m (4800 ft) for any

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surface deformities using a microscope as specified in paragraph 4.7.2.1.

4.7.2.14. Mechanical Interface. Mount head assembly on tape transport type for which head assembly is to perform. Observe and verify that mechanical interfaces are compatible with tape transport. Visually inspect for OEM conformance.

4.7.2.15. Azimuth Reproduce Output. Record and reproduce a tape according to procedures in 4.7.2.4 at bandedge and optimize the azimuth of the first track of the head stack, and log the output. Repeat procedure for remaining tracks while logging previously optimized tracks and verify no greater deviation than specified by this requirement.

4.7.2.16. Pole Tip Hardness. Using a sample from the exact lot of poletip material to be used in the head under test, perform hardness test according to one of the procedures listed in Table III, reference ASTM-E10-73 and ASTM E104-76. Log the results.

4.7.2.17. Test Data. Provide test data in accordance with OEM specifications and DD Form 1423.

4.7.2.18. Thermal Rate of Change. After head assembly has performed to paragraph 4.8.2.4, remove assembly from transport and insert in test chamber according to OEM specifications. Reinstall head assembly on tape transport and verify performance according to 4.7.2.4. Verify performance per 4.7.2.4 to this specification.

4.7.2.19. Certification Compliance. The manufacturer shall provide written certification of materials, dimensions, electrical characteristics, physical characteristics tests and workmanships, as required by Section 3, Requirements.

5.0. PREPARATION FOR DELIVERY

5.1. Packaging. Packaging shall be in accordance with the provisions of the contract.

5.2. Marking. The unit package, intermediate package and shipping container, shall be marked with the date of acceptance by the Government and any markings specified in the contract.

6.0. NOTES

6.1. Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type of tape head required must be specified.

c. Selection of applicable levels of packaging and packing.

d. Identification of agency of the U.S. Government (or contractor) which will perform first article testing and the number of first articles to be tested. If the bidding vendor can provide documentation substantiating previous qualification within the past 24 months from the date of this procurement on the specific part number(s) of head assemblies procured under this contract and the vendor certifies that no material changes or manufacturing processes changes have been made in the fabrication of the head since previous

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qualification, first article inspection may be waived after award of contract.

e. Manufacturer shall provide record or reproduce head electronic data sheets (see 3.2.5.2 and 3.2.5.3).

f. Applicable national stock number.

g. Marking requirements for packaging/packing.

h. Government furnished equipment (GFE), i.e., sample OEM head or tape transport, if required.

6.2. Definition of Terms Used. Terms used in this specification are as defined in IRIG-106-80 and IRIG-118-79.

6.3. Recycled, Virgin, and Reclaimed Materials. Reclaimed materials shall be used to maximum extent possible.

6.4. Qualified Products List. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on a Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the DOD tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The verifying activity responsible for the qualified products list is SM-ALC/PMZ, McClellan AFB CA 95652, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Application by Manufacturers for Inclusion of Federal Qualified Products Lists (QPLs)." (Copies of "Provisions Governing Application by Manufacturers for Inclusion on Federal Qualified Products Lists (QPLs)" may be obtained upon application to Director, Standardization Division, Federal Supply Service, General Services Administration, Washington DC 20406.

6.5. Definitive Specification Part Number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirements options available in this specification as follows:

Definitive Specification Part Number MXXXXX-XX-XX
Military Specification No._____
Classification Type No. (See 1.2)_____
Head Type No. (See Table I)_____

6.5.1. Sample Part Numbers

Characteristics	Part Number
Class Type I Head Type 9	M12345-0I-09
Class Type II Head Type 10	M12345-II-10

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Preparing Activity:

80

Review Activities:

Army CR
Navy EC

Civil Agency Interest:

GSA-FSS
NASA
NASA

User Activities;

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DOCUMENT IDENTIFIER AND TITLE
MIL-H-87969 Heads, Magnetic Tape, Instrumentation Irig Wide Band Group II

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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